Tayten Bennetsen

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EDUCATION

Rice University **Houston, Texas**
B.S. in Computer Science and B.A. in Cognitive Science Expected Graduation, May 2027

- **GPA:** 3.75/4.00
- **Concentrations:** DSA, System Design, Machine Learning, OOP, OS, Data Analysis, Embedded Programming, Networking

EXPERIENCE

Backend Engineer Intern | Voodoo — June 2025 – Sep 2025
Nomad the iOS social travel journal — Swift Concurrency | Python | AWS | Git | Google Maps API | Apify Scraper

- Collaborated with team of 4 on an iOS **travel-social** application that overlays social network driven recommendations for **50+ cities**.
- Built **SVD-based recommendation engine** to solve content discovery challenges, enabling personalized user-generated content feeds.
- Developed **16 core AWS Lambda** functions for back-end authentication, Maps/TikTok scraping, and CRUD, reducing server costs by **60%**.

AI Software Engineer Intern | Wostmann & Associates — May 2025 – Aug 2025
Agentic Procurement scraper — Python | Django | Docker | Claude MCP | TensorFlow | Redis | Playwright MCP

- Architected a **Dockerized Python/Django** agentic pipeline to scrape, classify, and rank opportunities, boosting lead discovery efficiency by **70%**.
- Integrated **neural network ranking logic** (TensorFlow + NLP heuristics) with **99.5%** relevance accuracy to surface relevant opportunities.
- Deployed automated scraping via Claude and Playwright MCP **containerized agentic pipelines**, scraping **120** websites from government, state, and private sectors into a **Postgres** database.

Founding Software Engineer | Signacer — Nov 2024 – Aug 2025
Athlete–fan connection app — Swift | SwiftUI | Firebase | Git | Swift Concurrency

- Led **Swift** development for a real-time athlete–fan engagement platform, implementing QR authentication, chat, and notification services.
- Architected **Firebase** back-end (Firestore, Cloud Functions, Swift Concurrency) to support scalable data storage, messaging, and async handling.
- Coordinated feature planning and sprint execution in an **Agile** workflow alongside co-founders to connect **400** onboarded users.

Software Engineer Intern | Wostmann & Associates — May 2024 – Aug 2024
Open source screen reader for visually impaired users — C++ | WinAPI | Microsoft UI Automation | Windows Speech API

- Designed **SightSpeak Reader** using C++, integrating WinAPI, Microsoft UI Automation, and Windows Speech API to enable real-time text-to-speech output and visual text highlighting, supporting over **4,000** Microsoft applications.
- Developed recursive BFS tree traversal for seamless navigation through UI elements, improving navigation speed by **80%** for beta users.
- Leveraged a **thread pool** and futures/promises for concurrent tasks, reducing text processing latency by **150ms** for smooth user interface.

PROJECTS

Full-Stack Project | Bunkmate — Sep 2024 – Present
Rice University centralized student housing and subletting platform — Next.js | TypeScript | Supabase | Tailwind CSS | Google Maps API

- Developed a **Next.js/TypeScript** application enabling Rice students to post, search, and manage sublets/off-campus housing.
- Implemented **Supabase Auth** with role-based access control to secure user data and restrict listings management with **6** authorization rules.
- Architected real-time database interactions in **Supabase** for favorites, user profiles, and listing updates, resulting in over **2000** users and **500+** listings.
- Developed as Tech Lead as part of **Rice Apps** student developer community, leading weekly agile sprints and scaling to nation wide service.

Embedded Software Project | Rice OwlSat — Oct 2024 – May 2025
NASA-sponsored embedded satellite software — CircuitPython | Raspberry Pi | I2C Sensors | PyCubed

- Developed **embedded software** in CircuitPython for Raspberry Pi 2350, reading sensor telemetry (IMU, GPS, temperature) over I2C and encoding data for ground-station transmission.
- Integrated sensor drivers and implemented fault-tolerant dual threaded data buffering, ensuring reliable operation under variable power conditions.
- Conducted **200+ hours of comprehensive testing** and finalize flight software architecture for the **NASA-sponsored** CubeSat project.

AI Full-Stack Project | Pillora – AI Powered Medication Management — Sep 2024 – Oct 2024
HackRice 2024 Winner — JavaScript | React | Node.js | MongoDB | Tailwind CSS | OpenAI | OpenFDA | Express

- Led a **team of 4** to build a cross-platform React/Node.js app focused on improving medication adherence for patients.
- Implemented **Retrieval-Augmented Generation (RAG)** by integrating OpenFDA API with **OpenAI**, enhancing chatbot accuracy by **35%** and reducing hallucinations by **90%**.
- Architected a scalable backend using **MongoDB** and Express, supporting up to **250** concurrent users with minimal latency.
- Awarded **2nd Place Overall** and **1st in Healthcare** at HackRice 2024.

TECHNICAL SKILLS

Languages | Python, Java, Go, C++, C, JavaScript, TypeScript, Swift, React, R, SQL, HTML, CSS, Rust
Programming Tools | AWS, Docker, Git, Firebase, Supabase, MongoDB, Postgres, Redis, OpenAI, Anthropic, Linux, Kubernetes
Frameworks/Libraries | LangChain, TensorFlow, Pandas, PyTorch, Django, Scikit-Learn, Next.js, Node.js, Django, JUnit